

Indian Institute of Technology Gandhinagar

BE623 — Biocomputing | Sem 1, 2026-2027

Git & GitHub Quickstart

Account setup, SSH keys, and terminal login/logout — supplement to Lab Session 3

Step 1 — Create a GitHub Account (Browser)

This is the only step done outside the terminal.

- Open a browser and go to github.com/join
- Enter a username, email, and password, and verify the email address.
- Use your institute or personal email. You will need to confirm it before continuing.

Step 2 — Generate an SSH Key (Terminal)

An SSH key proves your identity to GitHub without typing a password every time.

```
$ ssh-keygen -t ed25519 -C "your_email@example.com"
```

Press Enter to accept the default file location (`~/.ssh/id_ed25519`). When asked for a passphrase, set one. It protects your key if the shared machine is later reused by someone else.

NOTE: A key pair is created: `id_ed25519` (private, never shared) and `id_ed25519.pub` (public, safe to share).

Step 3 — Load the Key and Set Your Git Identity (Terminal)

`ssh-agent` holds your unlocked key in memory for the session so you are not prompted for the passphrase on every git command. While here, also set your name/email once per machine so commits are attributed to you.

```
$ eval "$(ssh-agent -s)"
$ ssh-add ~/.ssh/id_ed25519
$ git config --global user.name "Your Name"
$ git config --global user.email "your_email@example.com"
```

NOTE: Replace your username, email, name wherever necessary.

Expected: Identity added: `/home/yourusername/.ssh/id_ed25519`

Step 4 — Add Your Public Key to GitHub (Browser)

Print the public key and copy it — this is the only piece of information GitHub needs.

```
$ cat ~/.ssh/id_ed25519.pub
```

- Copy the full output (starts with `ssh-ed25519` and ends with your email).
- On GitHub: Settings > SSH and GPG keys > New SSH key.
- Paste the key, give it a title (e.g. "BE623 Lab Machine"), and click Add SSH key.

Step 5 — Log In: Verify From the Terminal

GitHub over SSH has no separate "login" step. A successful connection test is your login.

```
$ ssh -T git@github.com
```

Expected: *Hi <your-username>! You've successfully authenticated, but GitHub does not provide shell access.*

WHY: *This means git clone/pull/push now work with no username or password prompts.*

Step 6 — Log Out From the Terminal

SSH authentication is not a session like a website login, so there is no literal "logout" command. The equivalent is removing your key from the agent so it can no longer be used without the passphrase again.

```
$ ssh-add -D
$ eval "$(ssh-agent -k)"
```

The first command removes all keys from the running agent; the second stops the agent process itself. Your key files in ~/.ssh remain on disk. Nothing is deleted.

CAUTION

On a shared institute machine, always run Step 6 (log out) before you walk away. Leaving your key loaded in ssh-agent lets the next person who sits at that terminal push and pull as you, essentially giving them access to manage your repos.

Troubleshooting Guide

Problem	Cause	Solution
Permission denied (publickey)	Public key was not added to GitHub, or wrong key is loaded	Re-check Step 4; confirm with ssh-add -l
Agent has no identities / cannot connect to agent	ssh-agent was never started, key removed (Step 6), or this is a new terminal session	Redo Step 3 (eval + ssh-add) — the agent does not persist across logins
Asked for a GitHub username/password	Using an HTTPS remote URL instead of SSH	Use the SSH clone URL: git clone git@github.com:user/repo.git
gh: command not found	GitHub CLI is not installed and cannot be (if no sudo)	Expected for the lab computers. Use plain git and ssh as in this guide

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